



TECNOLÓGICO
NACIONAL DE MÉXICO®

INSTITUTO TECNOLÓGICO DE NUEVO LAREDO

U3 – PRACTICA 7: DHCP ON ROUTER c/VLANS #3

Equipo Simulado y Físico

**CONMUTACION Y ENRUTAMIENTO DE
REDES DE DATOS**

Desarrollado por:

GERARDO ALBERTO CRUZ ROJO - 19100163

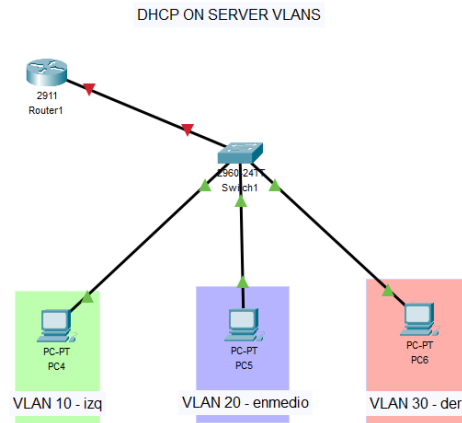
JUAN HERNANDEZ ROSAS - 19100199

DOCENTE: ING. ROBERTO DIAZ PUERTO

Fecha de Entrega: 15 DE NOVIEMBRE DE 2023



Primeramente configuramos las VLANS (10,20 y 30) en el switch:



```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname RT-DHCP-VLAN
RT-DHCP-VLAN(config)#vlan 10
RT-DHCP-VLAN(config-vlan)#name izq
RT-DHCP-VLAN(config-vlan)#vlan 20
RT-DHCP-VLAN(config-vlan)#name enmedio
RT-DHCP-VLAN(config-vlan)#vlan 30
RT-DHCP-VLAN(config-vlan)#name der
RT-DHCP-VLAN(config-vlan)#
```

Agregamos la VLAN 40 como native.

```
RT-DHCP-VLAN(config-vlan)#vlan 40
RT-DHCP-VLAN(config-vlan)#name native
RT-DHCP-VLAN(config-vlan)#
```

Asignamos el rango de interfaces para cada VLAN.

VLAN 10: fa0/1 – 0/9

```
RT-DHCP-VLAN(config-if-range)#int range fa0/1-9
RT-DHCP-VLAN(config-if-range)#switchport mode access
RT-DHCP-VLAN(config-if-range)#sw
RT-DHCP-VLAN(config-if-range)#switchport access vlan 10
RT-DHCP-VLAN(config-if-range)#
```

VLAN 20: fa0/10 – 0/19

```
RT-DHCP-VLAN(config-if-range)#int range fa0/10-19
RT-DHCP-VLAN(config-if-range)#sw
RT-DHCP-VLAN(config-if-range)#switchport mode access
RT-DHCP-VLAN(config-if-range)#
RT-DHCP-VLAN(config-if-range)#sw
RT-DHCP-VLAN(config-if-range)#switchport access vlan 20
RT-DHCP-VLAN(config-if-range)#
```

VLAN 30: fa0/20 – 0/24

```
RT-DHCP-VLAN(config-if-range)#int range fa0/20-24
RT-DHCP-VLAN(config-if-range)#sw
RT-DHCP-VLAN(config-if-range)#switchport mode access
RT-DHCP-VLAN(config-if-range)#sw
RT-DHCP-VLAN(config-if-range)#switchport access vlan 30
RT-DHCP-VLAN(config-if-range)#
```

Configuramos el switch modo Truncal para la interface gig0/1, también será la VLAN 40 (nativa).

```
RT-DHCP-VLAN(config-if-range)#int gi0/1
RT-DHCP-VLAN(config-if)#sw
RT-DHCP-VLAN(config-if)#switchport mode trunk
RT-DHCP-VLAN(config-if)#sw
RT-DHCP-VLAN(config-if)#switchport trunk native vlan 40
RT-DHCP-VLAN(config-if)#
```

Encapsulamiento de interface.

En el router realizamos la siguiente configuración:

VLAN 10:

```
Router(config)#int gi0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Router(config-subif)#
```

VLAN 20:

```
Router(config-subif)#int gi0/0.20
Router(config-subif)#encapsulation dot1Q
% Incomplete command.
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#
```

VLAN 30:

```
Router(config-subif)#int gi0/0.30
Router(config-subif)#encapsulation dot1Q
% Incomplete command.
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.3.1
% Incomplete command.
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
```

VLAN 40 (nativa):

```
Router(config-subif)#int gi0/0.40
Router(config-subif)#encapsulation dot1Q 40 native
Router(config-subif)#
```

Configuracion de DHCP mediante el Router.

Para la VLAN 10 (izq)

```
Router(config)#ip dhcp pool izq10
Router(dhcp-config)#default-router
^
% Invalid input detected at '^' marker.
Router(dhcp-config)#default-router 192.168.1.1
^
% Invalid input detected at '^' marker.
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#
```

Para VLAN 20 (en medio)

```
Router(dhcp-config)#ip dhcp pool enmedio
Router(dhcp-config)#default-router
% Incomplete command.
Router(dhcp-config)#default-router 192.168.2.1
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#
```

Para VLAN 30 (der)

```
Router(dhcp-config)#ip dhcp pool der
Router(dhcp-config)#default-router 192.168.3.1
^
% Invalid input detected at '^' marker.
Router(dhcp-config)#default-router 192.168.3.1
Router(dhcp-config)#network 192.168.3.0 255.255.255.0
Router(dhcp-config)#
```

Ahora en cada host de cada VLAN, ya debe tener un IP por DHCP, correspondiente a su segmento.

En la VLAN 10:

☒ DHCP	☐ Static	DHCP request successful.
IPv4 Address	192.168.1.2	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.1.1	
DNS Server	0.0.0.0	

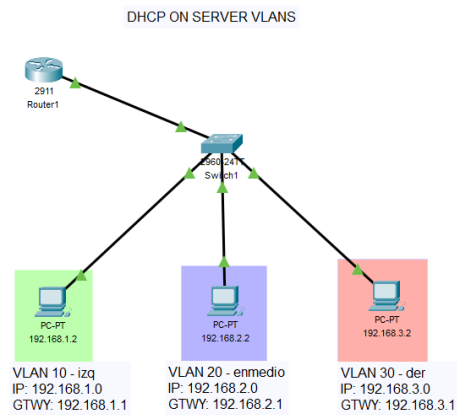
VLAN 20:

☒ DHCP	☐ Static	DHCP request successful.
IPv4 Address	192.168.2.2	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.2.1	
DNS Server	0.0.0.0	

VLAN 30:

☒ DHCP	☐ Static	DHCP request successful.
IPv4 Address	192.168.3.2	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.3.1	
DNS Server	0.0.0.0	

Diseño final:



Comprobación de conectividad entre distintas VLANs:

De 1.2 (Vlan 10) a 2.2 (Vlan 20)

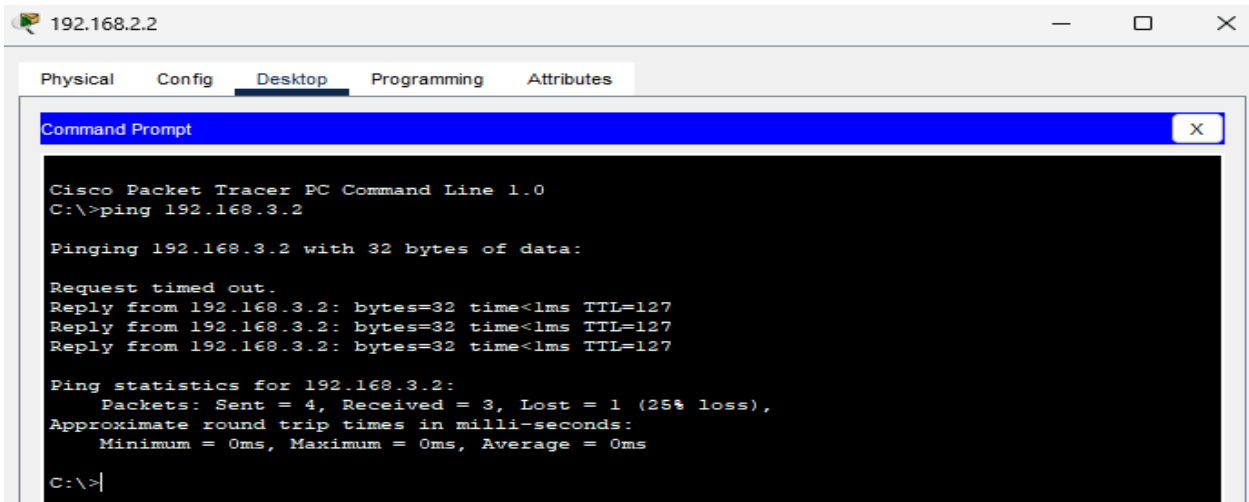
```
192.168.1.2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=4ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms
C:\>
```

De 2.2 (Vlan 20) a 3.2 (Vlan 30)



The image shows a Cisco Packet Tracer PC Command Line window for the IP address 192.168.2.2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt shows the execution of a ping command to 192.168.3.2. The output indicates a request timed out, followed by three successful replies from 192.168.3.2. The ping statistics show 4 packets sent, 3 received, and 1 lost (25% loss).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

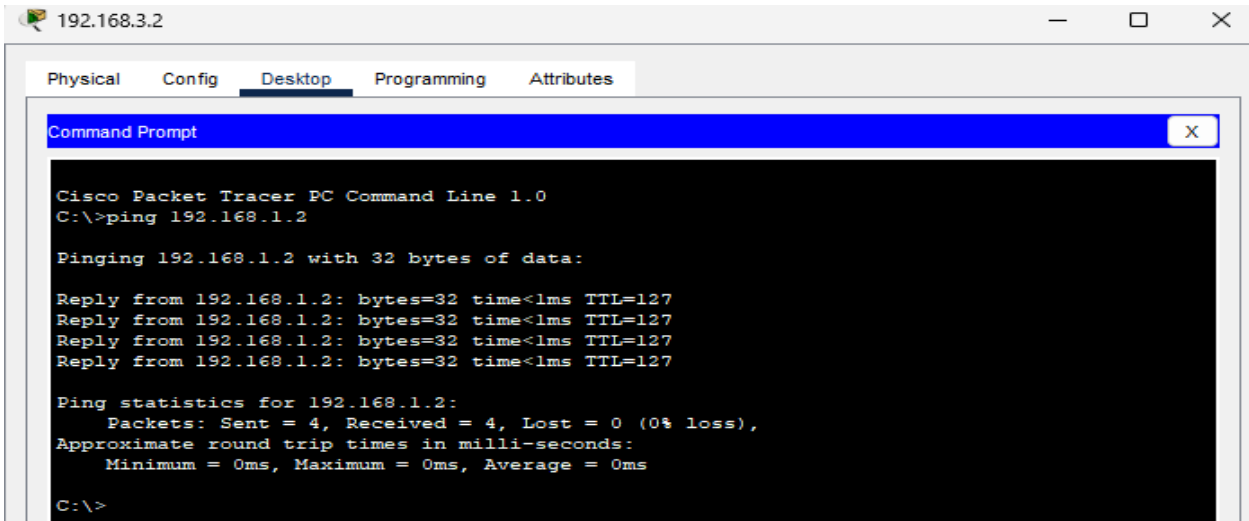
Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

De 3.2 (Vlan 30) a 1.2 (Vlan 10)



The image shows a Cisco Packet Tracer PC Command Line window for the IP address 192.168.3.2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt shows the execution of a ping command to 192.168.1.2. The output shows four successful replies from 192.168.1.2. The ping statistics show 4 packets sent, 4 received, and 0 lost (0% loss).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

EQUIPO FISICO:

CONFIGURACION **DHCP on Router c/VLANs #3**



Configuracion del router

```
COM6 - PuTTY
e to down
*Nov 17 16:26:58.223: %LINK-3-UPDOWN: Interface GigabitEthernet
e to up
*Nov 17 16:26:59.223: %LINEPROTO-5-UPDOWN: Line protocol on Int
ernet0/0, changed state to up
^
% Invalid input detected at '^' marker.

RT-DHCP(config-if)#interface gig0/0.10
RT-DHCP(config-subif)#encapsulation dot1Q 10
RT-DHCP(config-subif)#ip address 192.168.10.1 255.255.255.0
RT-DHCP(config-subif)#interface gig0/0.20
RT-DHCP(config-subif)#encapsulation dot1Q 20
RT-DHCP(config-subif)#ip address 192.168.20.1 255.255.255.0
RT-DHCP(config-subif)#interface gig0/0.30
RT-DHCP(config-subif)#encapsulation dot1Q 30
RT-DHCP(config-subif)#ip address 192.168.30.1 255.255.255.0
RT-DHCP(config-subif)#encapsulation dot1Q 40
RT-DHCP(config-subif)#encapsulation dot1Q 30
RT-DHCP(config-subif)#ip address 192.168.30.1 255.255.255.0
RT-DHCP(config-subif)#interface gig0/0.40
RT-DHCP(config-subif)#encapsulation dot1Q 40 native
RT-DHCP(config-subif)#ip address 192.168.40.1 255.255.255.0
RT-DHCP(config-subif)#
```

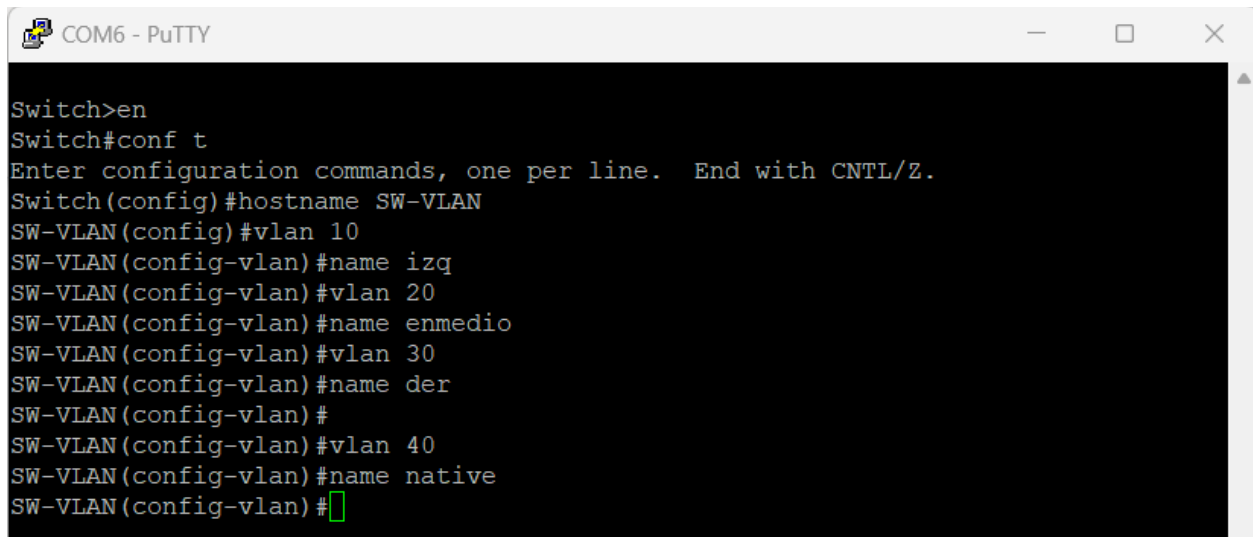

Configuramos el DHCP en el Router

 COM6 - PuTTY

```
RT-DHCP(config-subif)#ip address 192.168.10.1 255.255.255.0
RT-DHCP(config-subif)#interface gig0/0.20
RT-DHCP(config-subif)#encapsulation dot1Q 20
RT-DHCP(config-subif)#ip address 192.168.20.1 255.255.255.0
RT-DHCP(config-subif)#interface gig0/0.30
RT-DHCP(config-subif)#encapsulation dot1Q 30
RT-DHCP(config-subif)#ip address 192.168.30.1 255.255.255.0
RT-DHCP(config-subif)#encapsulation dot1Q 40
RT-DHCP(config-subif)#encapsulation dot1Q 30
RT-DHCP(config-subif)#ip address 192.168.30.1 255.255.255.0
RT-DHCP(config-subif)#interface gig0/0.40
RT-DHCP(config-subif)#encapsulation dot1Q 40 native
RT-DHCP(config-subif)#ip address 192.168.40.1 255.255.255.0
RT-DHCP(config-subif)#interface gig0/0.10
RT-DHCP(config-subif)#ip dhcp pool izq
RT-DHCP(config-subif)#ip dhcp pool izq10
RT-DHCP(dhcp-config)#default-router 192.168.10.1
RT-DHCP(dhcp-config)#network 192.168.10.0 255.255.255.0
RT-DHCP(dhcp-config)#ip dhcp pool enmedio
RT-DHCP(dhcp-config)#default-router 192.168.20.1
RT-DHCP(dhcp-config)#network 192.168.20.0 255.255.255.0
RT-DHCP(dhcp-config)#ip dhcp pool der
RT-DHCP(dhcp-config)#network 192.168.30.0 255.255.255.0
RT-DHCP(dhcp-config)#default-router 192.168.30.1
RT-DHCP(dhcp-config)#
RT-DHCP(dhcp-config)#
```

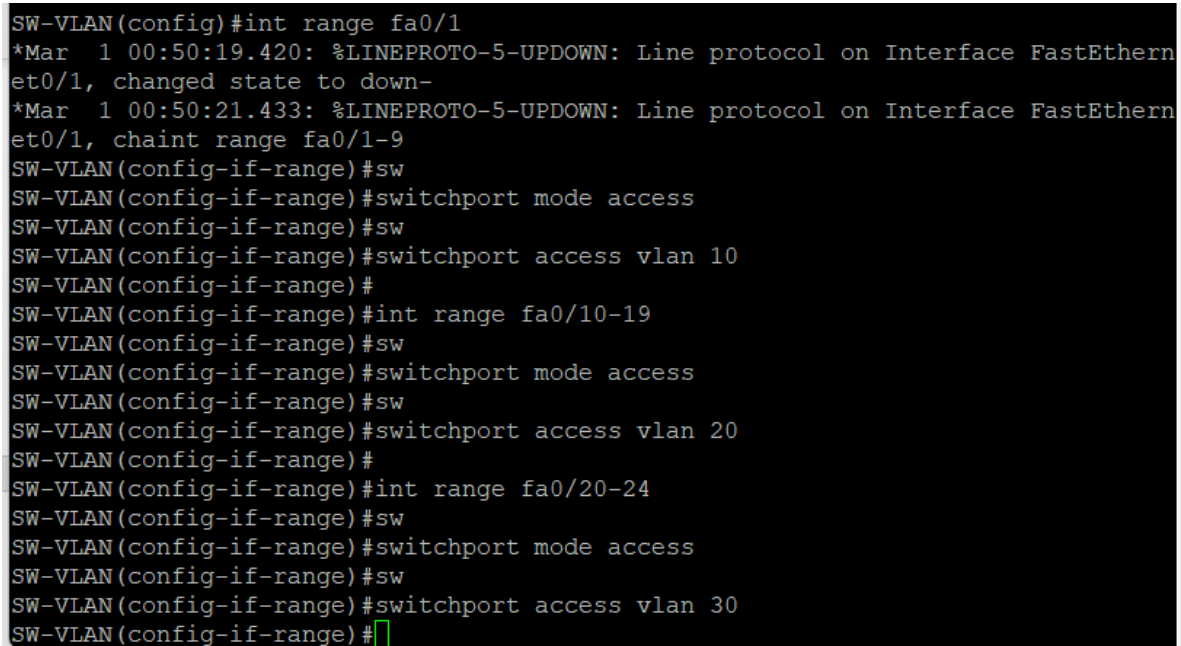

Configuramos el switch

Agregar VLANS



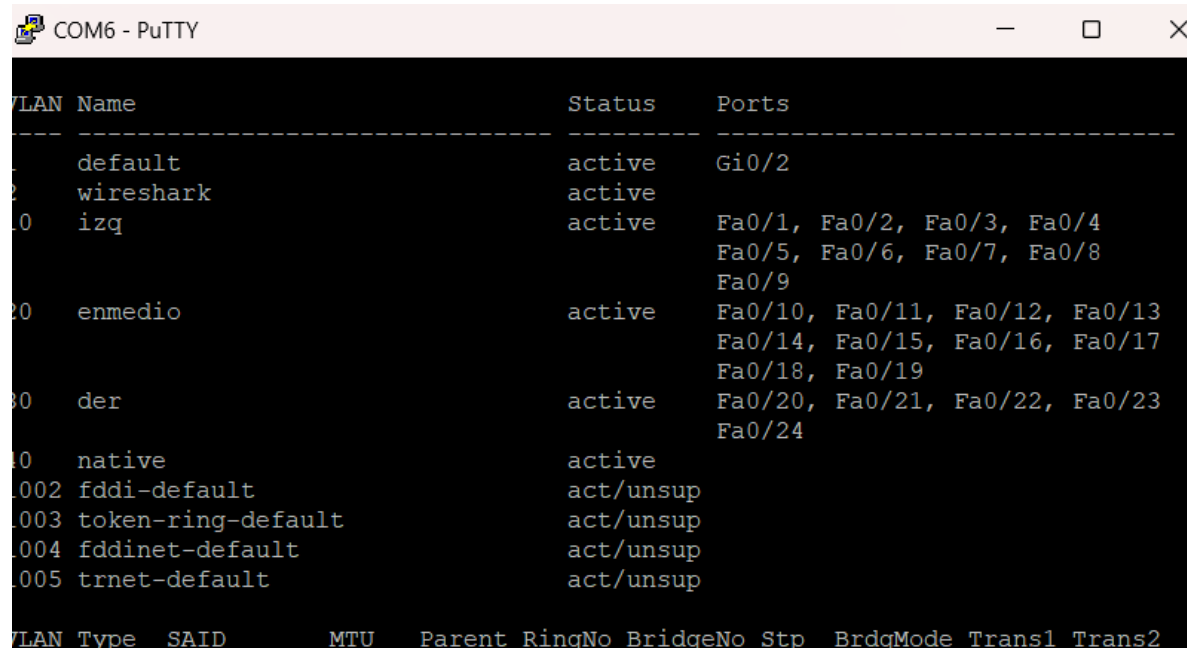
```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW-VLAN
SW-VLAN(config)#vlan 10
SW-VLAN(config-vlan)#name izq
SW-VLAN(config-vlan)#vlan 20
SW-VLAN(config-vlan)#name enmedio
SW-VLAN(config-vlan)#vlan 30
SW-VLAN(config-vlan)#name der
SW-VLAN(config-vlan)#
SW-VLAN(config-vlan)#vlan 40
SW-VLAN(config-vlan)#name native
SW-VLAN(config-vlan)#
```

Distribuimos puertos en las VLANS los rangos en mode Access cada uno



```
SW-VLAN(config)#int range fa0/1
*Mar  1 00:50:19.420: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down-
*Mar  1 00:50:21.433: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, chaint range fa0/1-9
SW-VLAN(config-if-range)#sw
SW-VLAN(config-if-range)#switchport mode access
SW-VLAN(config-if-range)#sw
SW-VLAN(config-if-range)#switchport access vlan 10
SW-VLAN(config-if-range)#
SW-VLAN(config-if-range)#int range fa0/10-19
SW-VLAN(config-if-range)#sw
SW-VLAN(config-if-range)#switchport mode access
SW-VLAN(config-if-range)#sw
SW-VLAN(config-if-range)#switchport access vlan 20
SW-VLAN(config-if-range)#
SW-VLAN(config-if-range)#int range fa0/20-24
SW-VLAN(config-if-range)#sw
SW-VLAN(config-if-range)#switchport mode access
SW-VLAN(config-if-range)#sw
SW-VLAN(config-if-range)#switchport access vlan 30
SW-VLAN(config-if-range)#
```

Comprobamos que se hicieron bien las configuraciones de las VLANs Y SUS rangos de puertos de cada una.

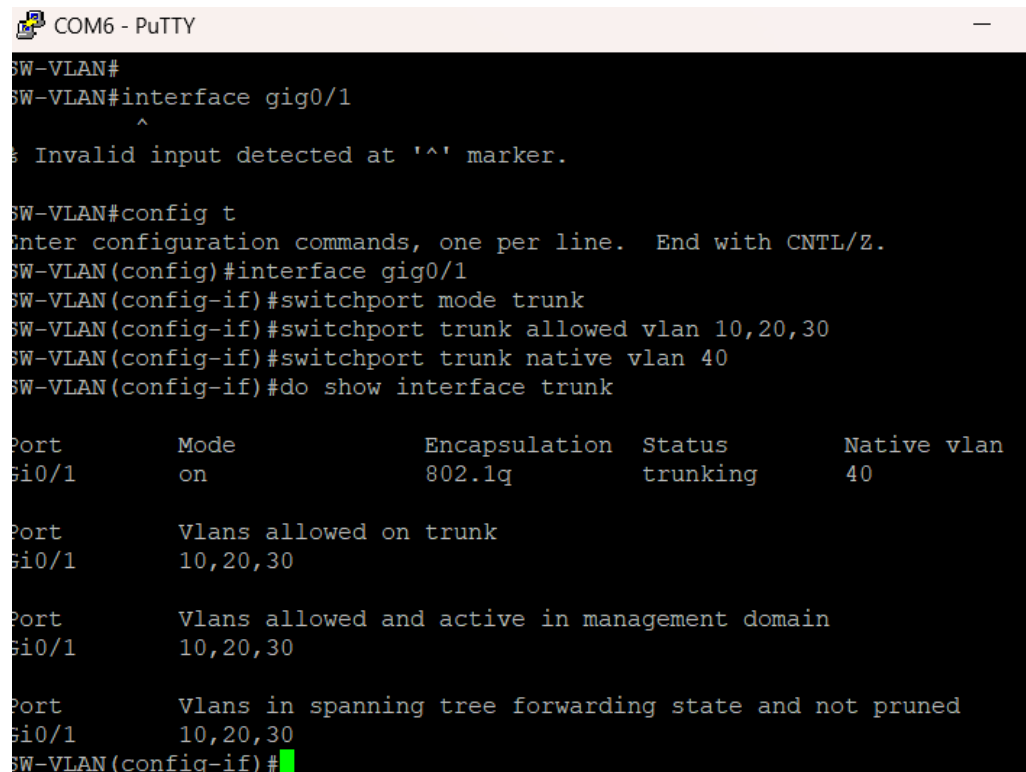


COM6 - PuTTY

VLAN	Name	Status	Ports
	default	active	Gi0/2
2	wireshark	active	
10	izq	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9
20	enmedio	active	Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19
30	der	active	Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
40	native	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
------	------	------	-----	--------	--------	----------	-----	----------	--------	--------

Configuramos el modo trunk en gig0/1 del extremo del switch al router y damos a conocer las VLANs que pasaran por el modo troncal.



COM6 - PuTTY

```
SW-VLAN#
SW-VLAN#interface gig0/1
^
% Invalid input detected at '^' marker.

SW-VLAN#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-VLAN(config)#interface gig0/1
SW-VLAN(config-if)#switchport mode trunk
SW-VLAN(config-if)#switchport trunk allowed vlan 10,20,30
SW-VLAN(config-if)#switchport trunk native vlan 40
SW-VLAN(config-if)#do show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/1	on	802.1q	trunking	40

Port	Vlans allowed on trunk
Gi0/1	10,20,30

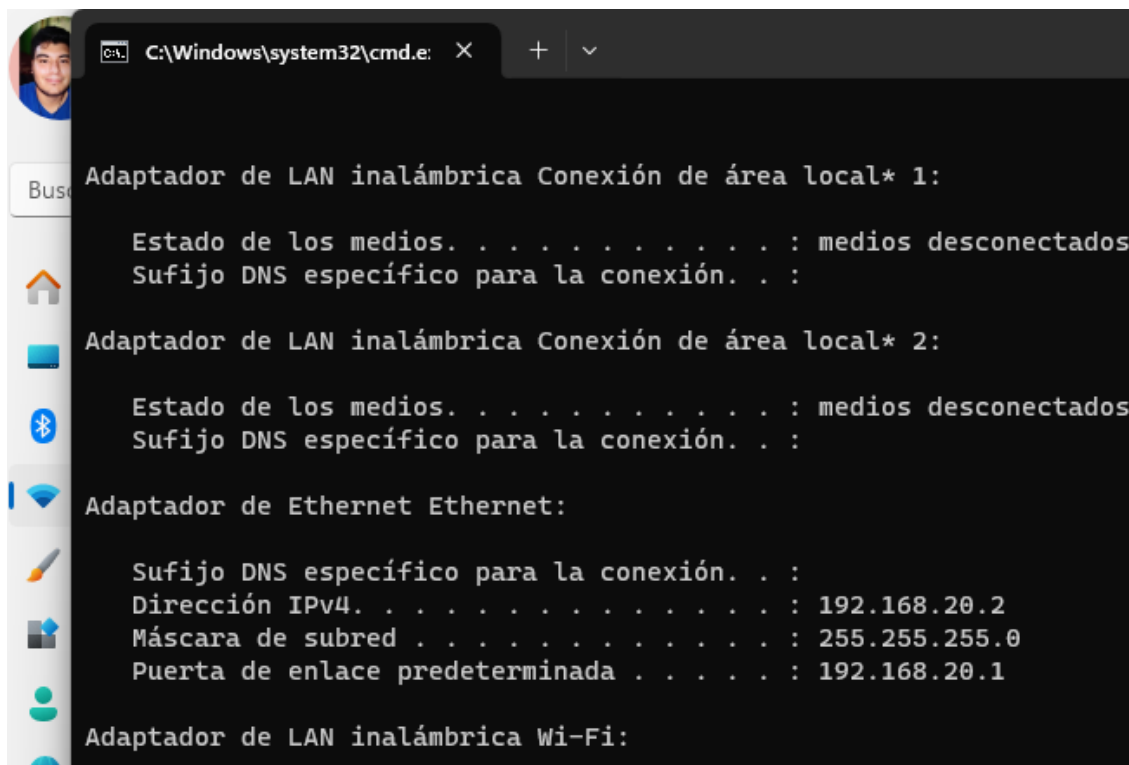
Port	Vlans allowed and active in management domain
Gi0/1	10,20,30

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/1	10,20,30

```
SW-VLAN(config-if)#
```

COMPROBACIONES DE CONECTIVIDAD

Asignación de Ip Vlan 20



```
C:\Windows\system32\cmd.e: X + v

Adaptador de LAN inalámbrica Conexión de área local* 1:

Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :

Adaptador de LAN inalámbrica Conexión de área local* 2:

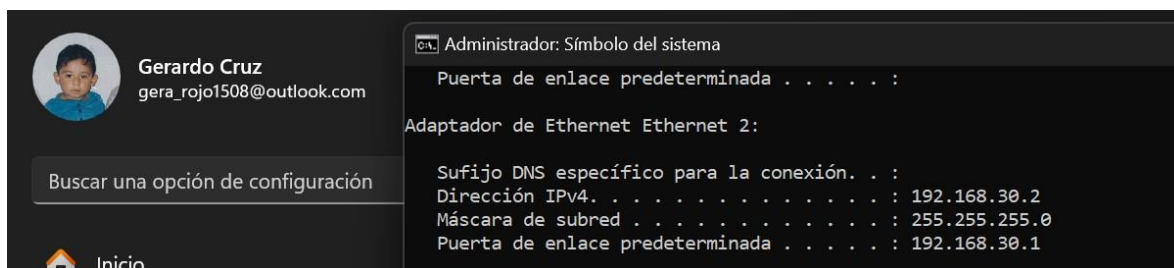
Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :

Adaptador de Ethernet Ethernet:

Sufijo DNS específico para la conexión. . :
Dirección IPv4. . . . . : 192.168.20.2
Máscara de subred . . . . . : 255.255.255.0
Puerta de enlace predeterminada . . . . . : 192.168.20.1

Adaptador de LAN inalámbrica Wi-Fi:
```

Asignación de Ip Vlan 30



```
Administrador: Símbolo del sistema

Puerta de enlace predeterminada . . . . . :

Adaptador de Ethernet Ethernet 2:

Sufijo DNS específico para la conexión. . :
Dirección IPv4. . . . . : 192.168.30.2
Máscara de subred . . . . . : 255.255.255.0
Puerta de enlace predeterminada . . . . . : 192.168.30.1
```

192.168.20.2 a 192.168.10.2

The screenshot shows the Windows Settings application with the 'Red e Internet > Ethernet' section selected. The left sidebar lists various settings categories, with 'Red e Internet' highlighted. The main pane displays network status for the Ethernet adapter, showing the IP address 192.168.20.1 and that the media is disconnected. A Command Prompt window is open in the foreground, showing the execution of a ping command from 192.168.20.2 to 192.168.10.2. The ping results show four successful responses with times ranging from 2ms to 4ms and a TTL of 127. The statistics indicate 4 packets sent and received with 0% loss.

```
C:\Windows\system32\cmd.e X + v
Puerta de enlace predeterminada . . . . . : 192.168.20.1

Adaptador de LAN inalámbrica Wi-Fi:

Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :

Adaptador de Ethernet Conexión de red Bluetooth:

Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :

C:\Users\juan7>ping 192.168.10.2

Haciendo ping a 192.168.10.2 con 32 bytes de datos:
Respuesta desde 192.168.10.2: bytes=32 tiempo=2ms TTL=127
Respuesta desde 192.168.10.2: bytes=32 tiempo=3ms TTL=127
Respuesta desde 192.168.10.2: bytes=32 tiempo=4ms TTL=127
Respuesta desde 192.168.10.2: bytes=32 tiempo=4ms TTL=127

Estadísticas de ping para 192.168.10.2:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
              (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 2ms, Máximo = 4ms, Media = 3ms

C:\Users\juan7>
```

Vlan 20 a Vlan30

PING 192.168.20.2 a 192.168.30.2

The screenshot shows the Windows Settings application with the 'Red e Internet > Ethernet' section selected. The left sidebar lists various settings categories, with 'Red e Internet' highlighted. The main pane displays network status for the Ethernet adapter, showing the IP address 192.168.20.1 and that the media is disconnected. A Command Prompt window is open in the foreground, showing the execution of a ping command from 192.168.20.2 to 192.168.30.2. The ping results show four successful responses with times ranging from 3ms to 4ms and a TTL of 127. The statistics indicate 4 packets sent and received with 0% loss.

```
C:\Windows\system32\cmd.e X + v
Puerta de enlace predeterminada . . . . . : 192.168.20.1

Adaptador de LAN inalámbrica Wi-Fi:

Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :

Adaptador de Ethernet Conexión de red Bluetooth:

Estado de los medios. . . . . : medios desconectados
Sufijo DNS específico para la conexión. . :

C:\Users\juan7>ping 192.168.30.2

Haciendo ping a 192.168.30.2 con 32 bytes de datos:
Respuesta desde 192.168.30.2: bytes=32 tiempo=4ms TTL=127
Respuesta desde 192.168.30.2: bytes=32 tiempo=4ms TTL=127
Respuesta desde 192.168.30.2: bytes=32 tiempo=3ms TTL=127
Respuesta desde 192.168.30.2: bytes=32 tiempo=4ms TTL=127

Estadísticas de ping para 192.168.30.2:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
              (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 3ms, Máximo = 4ms, Media = 3ms

C:\Users\juan7>
```